

```
/* =====  
ORACLE SYNONYM COMPLETE PRACTICE SCRIPT  
Scenario: Telecom Billing System  
===== */
```

```
/* =====  
1. CLEAN OLD OBJECTS  
===== */
```

```
BEGIN  
    EXECUTE IMMEDIATE 'DROP SYNONYM syn_subscribers';  
EXCEPTION WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN  
    EXECUTE IMMEDIATE 'DROP SYNONYM syn_call_records';  
EXCEPTION WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN  
    EXECUTE IMMEDIATE 'DROP VIEW vw_active_subscribers';  
EXCEPTION WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN  
    EXECUTE IMMEDIATE 'DROP PROCEDURE prc_get_subscriber_bill';  
EXCEPTION WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN  
    EXECUTE IMMEDIATE 'DROP TABLE call_records CASCADE CONSTRAINTS';  
EXCEPTION WHEN OTHERS THEN NULL;  
END;  
/
```

```
BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE subscribers CASCADE CONSTRAINTS';
EXCEPTION WHEN OTHERS THEN NULL;
END;
/
```

```
/* =====
2. CREATE BASE TABLES
===== */
```

```
CREATE TABLE subscribers
(
    subscriber_id NUMBER PRIMARY KEY,
    msisdn        VARCHAR2(20) UNIQUE NOT NULL,
    customer_name  VARCHAR2(100),
    city          VARCHAR2(50),
    status         VARCHAR2(20),
    activation_date DATE
);
```

```
CREATE TABLE call_records
(
    call_id        NUMBER PRIMARY KEY,
    subscriber_id  NUMBER,
    call_type      VARCHAR2(20),
    call_start     DATE,
    duration_sec   NUMBER,
    charge_amount  NUMBER(10,2),
    CONSTRAINT fk_call_subscriber
    FOREIGN KEY (subscriber_id)
    REFERENCES subscribers(subscriber_id)
);
```

```
/* =====
3. POPULATE SAMPLE DATA
===== */
```

INSERT INTO subscribers VALUES

(1, '959400000001', 'Aung Aung', 'Yangon', 'ACTIVE', DATE '2024-01-01');

INSERT INTO subscribers VALUES

(2, '959400000002', 'Mg Mg', 'Mandalay', 'ACTIVE', DATE '2024-01-05');

INSERT INTO subscribers VALUES

(3, '959400000003', 'Hla Hla', 'Naypyitaw', 'INACTIVE', DATE '2024-02-01');

INSERT INTO subscribers VALUES

(4, '959400000004', 'Su Su', 'Yangon', 'ACTIVE', DATE '2024-02-10');

INSERT INTO subscribers VALUES

(5, '959400000005', 'Kyaw Kyaw', 'Bago', 'SUSPENDED', DATE '2024-03-01');

INSERT INTO call_records VALUES

(101, 1, 'VOICE', DATE '2024-04-01', 120, 150.00);

INSERT INTO call_records VALUES

(102, 1, 'SMS', DATE '2024-04-02', 0, 15.00);

INSERT INTO call_records VALUES

(103, 2, 'DATA', DATE '2024-04-03', 500, 400.00);

INSERT INTO call_records VALUES

(104, 4, 'VOICE', DATE '2024-04-04', 300, 350.00);

INSERT INTO call_records VALUES

(105, 4, 'DATA', DATE '2024-04-05', 700, 600.00);

COMMIT;

/* =====

4. CREATE VIEW

===== */

CREATE OR REPLACE VIEW vw_active_subscribers

AS

```

SELECT
    subscriber_id,
    msisdn,
    customer_name,
    city,
    activation_date
FROM subscribers
WHERE status = 'ACTIVE';

```

```

/* =====
5. CREATE PROCEDURE
===== */

```

```

CREATE OR REPLACE PROCEDURE prc_get_subscriber_bill
(
    p_subscriber_id IN NUMBER
)
AS
    v_total_bill NUMBER(10,2);
BEGIN
    SELECT NVL(SUM(charge_amount), 0)
    INTO v_total_bill
    FROM call_records
    WHERE subscriber_id = p_subscriber_id;

    DBMS_OUTPUT.PUT_LINE('Subscriber ID: ' || p_subscriber_id);
    DBMS_OUTPUT.PUT_LINE('Total Bill: ' || v_total_bill);
END;
/

```

```

/* =====
6. CREATE PRIVATE SYNONYMS
===== */

```

```

CREATE SYNONYM syn_subscribers
FOR subscribers;

```

```

CREATE SYNONYM syn_call_records

```

FOR call_records;

CREATE SYNONYM syn_active_subscribers
FOR vw_active_subscribers;

CREATE SYNONYM syn_get_bill
FOR prc_get_subscriber_bill;

/* =====
7. TEST SYNONYM FOR TABLE SELECT
===== */

```
SQL> SELECT *  
FROM syn_subscribers;
```

```
  2  
SUBSCRIBER_ID MSISDN  
-----  
CUSTOMER_NAME  
-----  
CITY STATUS  
-----  
ACTIVATIO  
-----  
1 9594000000001  
Aung Aung  
Yangon ACTIVE  
01-JAN-24
```

```
SUBSCRIBER_ID MSISDN  
-----  
CUSTOMER_NAME  
-----  
CITY STATUS  
-----  
ACTIVATIO  
-----  
2 9594000000002  
Mg Mg  
Mandalay ACTIVE  
05-JAN-24
```

```
SQL> SELECT *
FROM syn_call_records;
```

CALL_ID	SUBSCRIBER_ID	CALL_TYPE	CALL_STAR	DURATION_SEC
101	1	VOICE	01-APR-24	120
102	1	SMS	02-APR-24	0
103	2	DATA	03-APR-24	500
104	4	VOICE	04-APR-24	300
105	4	DATA	05-APR-24	700

```
/* =====
```

8. TEST SYNONYM WITH WHERE CONDITION

```
===== */
```

```

SQL> SELECT subscriber_id,
        msisdn,
        customer_name,
        city
FROM syn_subscribers
WHERE city = 'Yangon';
  2      3      4      5      6
SUBSCRIBER_ID MSISDN
-----
CUSTOMER_NAME
-----
CITY
-----
                1 9594000000001
Aung Aung
Yangon

                4 9594000000004
Su Su
Yangon

```

/* =====

9. TEST SYNONYM WITH JOIN

===== */

```
SQL> SELECT s.customer_name,
           s.msisdn,
           c.call_type,
           c.duration_sec,
           c.charge_amount
FROM syn_subscribers s
JOIN syn_call_records c
ON s.subscriber_id = c.subscriber_id
WHERE s.status = 'ACTIVE';
  2    3    4    5    6    7    8    9
CUSTOMER_NAME
```

MSISDN	CALL_TYPE	DURATION_SEC	CHARGE_AMOUNT
Aung Aung 9594000000001	VOICE	120	150

Aung Aung 9594000000001	SMS	0	15
----------------------------	-----	---	----

Mg Mg 9594000000002	DATA	500	400
------------------------	------	-----	-----

CUSTOMER_NAME

MSISDN	CALL_TYPE	DURATION_SEC	CHARGE_AMOUNT
Su Su 9594000000004	VOICE	300	350

Su Su 9594000000004	DATA	700	600
------------------------	------	-----	-----

/* =====

10. TEST SYNONYM FOR VIEW

===== */


```

SQL> SELECT *
FROM syn_active_subscribers;
  2
SUBSCRIBER_ID MSISDN
-----
CUSTOMER_NAME
-----
CITY                                ACTIVATIO
-----
      1 9594000000001
Aung Aung
Yangon                                01-JAN-24

      2 9594000000002
Mg Mg
Mandalay                             05-JAN-24

SUBSCRIBER_ID MSISDN
-----
CUSTOMER_NAME
-----
CITY                                ACTIVATIO
-----

      4 9594000000004
Su Su
Yangon                                10-FEB-24

```

```

/* =====
11. TEST SYNONYM FOR PROCEDURE
===== */

```

```

SQL> SET SERVEROUTPUT ON;
SQL> EXEC syn_get_bill(1);
Subscriber ID: 1
Total Bill: 165

PL/SQL procedure successfully completed.

SQL> EXEC syn_get_bill(4);
Subscriber ID: 4
Total Bill: 950

PL/SQL procedure successfully completed.

```

```
/* =====
12. TEST INSERT USING SYNONYM
===== */
```

```
INSERT INTO syn_subscribers
VALUES
(6, '959400000006', 'Mya Mya', 'Yangon', 'ACTIVE', DATE '2024-03-15');

COMMIT;
```

```
SQL> SELECT *
FROM syn_subscribers
WHERE subscriber_id = 6;
      2      3
SUBSCRIBER_ID MSISDN
-----
CUSTOMER_NAME
-----
CITY                                     STATUS
-----
ACTIVATIO
-----
                6 9594000000006
Mya Mya
Yangon                                     ACTIVE
15-MAR-24
```

```
/* =====
13. TEST UPDATE USING SYNONYM
===== */
```

```
UPDATE syn_subscribers
SET status = 'INACTIVE'
WHERE subscriber_id = 6;

COMMIT;
```

```
SQL> SELECT *
FROM syn_subscribers
WHERE subscriber_id = 6;
      2      3
SUBSCRIBER_ID MSISDN
```

```
-----
CUSTOMER_NAME
-----
CITY                                STATUS
-----
ACTIVATIO
-----
      6 9594000000006
Mya Mya
Yangon                                INACTIVE
15-MAR-24
```

```
/* =====
```

```
14. TEST DELETE USING SYNONYM
```

```
===== */
```

```
DELETE FROM syn_subscribers
WHERE subscriber_id = 6;
```

```
COMMIT;
```

```
SQL> SELECT *
FROM syn_subscribers
WHERE subscriber_id = 6;
      2      3
no rows selected
```

```
/* =====
```

```
15. CHECK CREATED SYNONYMS
```

```
===== */
```

```
SQL> SELECT synonym_name,
        table_owner,
        table_name
FROM user_synonyms
ORDER BY synonym_name;
```

```
2      3      4      5
SYNONYM_NAME
```

```
-----
TABLE_OWNER
```

```
-----
TABLE_NAME
```

```
-----
CATALOG
SYS
CATALOG
```

```
COL
SYS
COL
```

```
SYNONYM_NAME
```

```
-----
TABLE_OWNER
```

```
-----
TABLE_NAME
```

```
-----
PRODUCT_USER_PROFILE
SYSTEM
SQLPLUS_PRODUCT_PROFILE
```

```
PUBLICSYN
SYS
```

```
/* =====
```

16. PUBLIC SYNONYM EXAMPLE

```
=====
```

Note:

Public synonym requires CREATE PUBLIC SYNONYM privilege.

Usually DBA creates it.

```
===== */
```

```
CREATE PUBLIC SYNONYM pub_subscribers
FOR subscribers;
```

```

SQL> SELECT *
FROM pub_subscribers;
  2
SUBSCRIBER_ID MSISDN
-----
CUSTOMER_NAME
-----
CITY STATUS
-----
ACTIVATIO
-----
      1 9594000000001
Aung Aung
Yangon ACTIVE
01-JAN-24

SUBSCRIBER_ID MSISDN
-----
CUSTOMER_NAME
-----
CITY STATUS
-----
ACTIVATIO
-----
      2 9594000000002
Mg Mg
Mandalay ACTIVE
05-JAN-24

```

```

/* =====

```

17. CROSS-SCHEMA SYNONYM EXAMPLE

```

=====

```

Example:

BILLING_USER owns the table.

CRM_USER wants to access it.

```

===== */

```

```

-- Step 1: Login as BILLING_USER

```

```

-- GRANT SELECT, INSERT, UPDATE, DELETE

```

```

-- ON subscribers

```

```
-- TO crm_user;
```

```
-- Step 2: Login as CRM_USER
```

```
-- CREATE SYNONYM crm_subscribers
```

```
-- FOR billing_user.subscribers;
```

```
-- SELECT *
```

```
-- FROM crm_subscribers;
```

```
/* =====  
18. DROP SYNONYM  
===== */
```

```
-- DROP SYNONYM syn_subscribers;
```

```
-- DROP SYNONYM syn_call_records;
```

```
-- DROP SYNONYM syn_active_subscribers;
```

```
-- DROP SYNONYM syn_get_bill;
```

```
-- DROP PUBLIC SYNONYM pub_subscribers;
```

```
/* =====  
19. COMMON ERRORS  
=====
```

Error 1:

ORA-00942: table or view does not exist

Possible reasons:

- Base table does not exist
- No privilege on base table
- Wrong schema name
- Synonym points to invalid object

Error 2:

ORA-01031: insufficient privileges

Possible reasons:

- User has no CREATE SYNONYM privilege
- User has no CREATE PUBLIC SYNONYM privilege

Error 3:

ORA-01775: looping chain of synonyms

Possible reason:

- Synonym points to another synonym repeatedly

===== */

/* =====

20. KEY NOTES

=====

1. Synonym is an alias for a database object.
2. Synonym does not store data.
3. Synonym can point to table, view, sequence, procedure, function, package, or materialized view.
4. Private synonym belongs to one schema.
5. Public synonym is available to all users.
6. Synonym simplifies object access.
7. Synonym hides schema names.
8. Synonym does not replace privileges.
9. User still needs permission on the base object.
10. If the base object is dropped, synonym becomes invalid.

===== */